Technical supplement: the validated Bellman/domain-wall certificate

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Abstract

This supplement specifies the computer-assisted input to the exact I_{3322} wall theorem. It records the characteristic map, hyperbolic plateau, analytic unstable-manifold enclosure, Miranda shooting box, exact contact covariance, global graph covers, boundary and exterior exclusions, the spatial-realization guards, the two-sided dimension law and its blind reconstruction, and the boundary between certified theorem and numerical conjecture. All inequalities that carry the theorem are interval-certified; symbolic identities are exact.

1 Arithmetic model and theorem boundary

Rigorous inequalities use Arb real balls through `python-flint`; exact identities use SymPy. A displayed decimal is descriptive only. A numerical gate passes only when an interval excludes zero with the registered sign.

The computer-assisted proposition supplies a positive concave $F : [-1, 1] \to (0, \infty)$, a strictly increasing predecessor map P, and

$$q_* \in [0.2508753845139765355 14, 0.2508753845139765364 86]$$

such that, with $b(x) = \sqrt{1 - x^2}/2$, $p(x) = b(x)^2/F(x)$, and $d(x, u) = xu + (x - u)/2 - 1$,

$$p(x) + F(u) \le q_* - d(x, u), \tag{S1}$$

with equality at the unique point $x = P(u)$. It also produces finite tensor-product strategies approaching q_*. The operator certificate and finite-dimensional nonattainment then proceed analytically in the main paper.

2 Characteristic map

For a state (x, y, r) define $s_x = \sqrt{1 - x^2}$, $s_y = \sqrt{1 - y^2}$ and

$$D = xy + \frac{x - y}{2} - 1, \qquad v = \frac{2(q - D - s_x/(2r))}{s_y},$$

$$z = \frac{1}{2}\left(\frac{(1 - 2x) + 2yv/s_y}{v^2} - 1\right).$$

The characteristic recurrence is

$$\mathcal{M}_q(x, y, r) = (y, z, v). \tag{S2}$$

It is obtained from equality and stationarity in (S1). Direct symbolic substitution verifies the recurrence and its exact reversal symmetry. The selected orbit leaves a positive fixed plateau, hits a two-equation reflection section, and returns along the reflected branch.

3 Plateau and analytic tail

The fixed point (C, C, R) belongs to the exact algebraic branch

$$R = \frac{\sqrt{1 - C^2}(2C - 1)}{(1 - C)(2C + 1)}, \qquad q = \frac{4C^4 - 5C^2 + 2}{4C^2 - 1}, \quad \frac{\sqrt{3}}{2} < C < 1. \tag{S3}$$

Exact elimination isolates the branch and proves the relevant root count. At the selected point $D\mathcal{M}_q$ has two stable eigenvalues and one unstable eigenvalue. Their reference midpoints are

$$5.8379840653\ldots, \qquad 0.8603760495\ldots, \qquad 0.1473755392\ldots.$$

The interval certificate, not these rounded values, separates them from the unit circle.

The local unstable curve $\Pi(t)$ is normalized by

$$\mathcal{M}_q(\Pi(t)) = \Pi(\mu t). \tag{S4}$$

A high-order polynomial approximation is corrected by a graph transform on a complex ball. The certified data are summarized in Table 1.

quantity	certified bound
complex boundary tiles	400
real radicand margin	> 0.22545
denominator margin	> 1.0726
graph-transform contraction	< 0.180945557
adapted-coordinate correction	$< 1.859 \times 10^{-26}$
original-coordinate correction	$< 2.213 \times 10^{-25}$

Table 1: Analytic unstable-manifold enclosure.

4 Validated shooting zero

The unknowns are the plateau coordinate C and unstable parameter t. After four and five iterates, the two residuals are

$$h_1 = \texttt{crossing}[0] + \texttt{crossing}[1], \qquad h_2 = \texttt{after}[2] - 1/\texttt{crossing}[2]. \tag{S5}$$

They encode reflection of the position and amplitude. A preconditioned Miranda argument validates a zero in

$$C = 0.87827294518081245 21 \pm 4.86 \times 10^{-20},$$

$$t = 0.0037582873342893243 \pm 4.36 \times 10^{-20}.$$

Every opposite face has the required strict sign after local-manifold and propagation errors are included. The two tail allowances are below 1.9×10^{-23} and 9.95×10^{-25}. Propagating the box through (S3) gives the enclosure of q_*. Miranda supplies existence; branch isolation and the strict graph below supply the uniqueness used in the Bellman construction.

5 Exact contact covariance

Let the one-form on the Bellman equality surface be

$$\beta = dF(x) - (1/2 - y)\,dx.$$

An exact symbolic calculation yields

$$\mathcal{M}_q^* \beta = \frac{1}{v^2}\beta. \tag{S6}$$

All coefficient residuals reduce identically to zero. Since the local unstable tangent is in $\ker \beta$, (S6) transports the envelope relation along the certified orbit. This identity is the exact bridge from the shooting recurrence to the scalar Bellman function; it is not inferred from a plot or interpolant.

6 Global graph and endpoints

The central orbit is covered at 300-bit precision by 8,192 local plateau tiles and four central pieces of 32,768 tiles each. Every projected derivative box has the registered strict sign, every pivot is positive, and there are no failed tiles. Reflection supplies the opposite half. Thus the characteristic curve is single-valued and the predecessor P is strictly increasing.

An earlier six-forward-piece parametrization failed. The accepted cover is plateau-to-section plus reflection. Both facts remain in the machine receipt to prevent the corrected theorem from silently inheriting the rejected coordinate convention.

The active right wing begins at the unique root in a width-2×10^{-14} bracket around

$$t = -0.003719358976358651\ldots.$$

The root residual has derivative below -34.725. Two pieces of 32,768 boxes certify the graph to the endpoint, whose predecessor is near 0.898116482394039. Reflection supplies the left wing.

The inactive exterior uses 32,768 boxes. Successor monotonicity and the stationarity-target derivative exclude every additional contact; the reconstructed F has lower bound 0.2104. A first attempted guard based on a strict line derivative failed at the calibrated endpoint. The theorem uses the repaired successor-monotonicity guard, and the receipt retains the failed test.

7 Assembly of the Bellman proposition

The exact covariance, strict graph signs, endpoint wings, and inactive exclusion show that $F > 0$, that its contact slope is monotone, and hence that F is concave. The contact tangent gives (S1) globally, with exactly one contact for each u.

Hyperbolicity supplies square-summable plateau tails. Truncating the two-sided orbit and normalizing gives finite tensor-product strategies whose boundary errors tend to zero. Their Bell values converge to q_*. The analytic positive-operator decomposition proves the matching commuting upper bound.

8 Equality audit and independent reconstruction

Equality in the three positive remainders forces two decreasing bijections of the same finite ordered support. They must be the same reversal. Eliminating the resulting amplitude ratio gives

$$q_* \le -s + \sqrt{s} \le \frac{1}{4},$$

contradicting the lower endpoint of the certified interval. This argument has an independently written exact implementation based on rank tables and a separate symbolic elimination.

The release verifier freezes hashes, checks semantic gates, and can rerun all engines. The production path is deterministic replay. A second path uses `mpmath.iv`, a locally implemented rectangular complex-interval layer, and no imports from the Arb/FLINT production modules. It passes 18,000 rational-enclosure checks and independently reconstructs the plateau, degree-12 invariant series, analytic graph transform, Miranda zero, central graph, endpoint wing, and inactive exterior. All eight registered gates pass. Its directed shooting interval is

$$[0.25087538451397653551479341100689144 95\ldots,\\ 0.2508753845139765357198798108850040972\ldots],$$

which overlaps the production Arb enclosure; both endpoints round to the displayed 18-place value. Its analytic-tail allowance is $2.2592334301979876 \times 10^{-25}$, only $1.020883\ldots$ times the production allowance. This is method-independent internal reconstruction, not a claim about physical enactment or a substitute for the analytic proof.

9 Spatial realization audit

The sharpness certificate reconstructs one bi-infinite profile c_j and a positive geometrically decaying eigenvector $\lambda \in \ell^2(\mathbb{Z})$ satisfying $H(c)\lambda = q_*\lambda$. This is stronger than the finite truncation statement used in the original release. Installing the Pál–Vértesi alternating rank-one blocks directly on the two perfect matchings of $\mathbb{Z}$, and using

$$\psi = \sum_{j\in\mathbb{Z}} \lambda_j e_j \otimes e_j,$$

produces a normal spatial strategy with Bell value $\langle \lambda, H(c)\lambda\rangle = q_*$. The expansion is absolutely convergent: diagonal terms are controlled by $\|\lambda\|_2^2$ and neighbor terms by Cauchy–Schwarz.

The local Bell-to-Jacobi identity has two new guards. The production guard uses exact rational unit-circle points on twelve open and twelve endpoint-free fixtures; all identities vanish, while deleting one alternating receiver produces a nonzero residual in every control. The independent guard expands the smallest periodic carrier containing both matching parities and computes the polynomial remainder modulo only $s_j^2 + c_j^2 - 1$; the Bell residual and all projection residuals are zero. Neither engine imports the interval or nonattainment implementation.

10 Quantitative truncation rate

For the principal section $I_L = \{-L, \ldots, L\}$, normalized mass $S_L = \sum_{j\in I_L} \lambda_j^2$, and compressed Bell value v_L, summing the exact wall eigenvalue equation gives

$$q_* - v_L = \frac{h_{-L}\lambda_{-L-1}\lambda_{-L} + h_{L+1}\lambda_L\lambda_{L+1}}{S_L}.$$

An independent symbolic reconstruction verifies this identity, detects both missing-boundary controls, derives the plateau ratio formula from the stationary equations, and encloses

$$R \in [1.078092050802091, 1.078092050802094],\\ \log R \in [0.07519285919570098, 0.07519285919570368].$$

Together with the certified analytic tails, this proves

$$\lim_{L\to\infty} \frac{-\log(q_* - v_L)}{2L+1} = \log R$$

and therefore, for the unrestricted optimum Q_d allowing binary POVMs,

$$0 < q_* - Q_d \le \exp[-d\log R + O(1)].$$

This is an achievability upper bound on the dimension needed for a specified accuracy. It is not a device-independent dimension lower bound.

Numerical illustration. For aligned centered finite carriers of dimensions 31, 63, 127, 191, and 255, the gaps are respectively approximately

$$3.8267\times10^{-4},\ 2.4605\times10^{-5},\ 1.8872\times10^{-7},\ 1.5326\times10^{-9},\ 1.2458\times10^{-11}.$$

Their successive exponential rates converge to $0.07519366\ldots$, while the proved exponent is $\log R = 0.07519286\ldots$. These values illustrate the theorem and were not used to infer its exponent. The exact finite-support reversal alone remains unstable quantitatively; the necessity proof instead uses robust remainder localization and finite-rank packet transport.

11 Quantitative dimension necessity

Let Q_d allow arbitrary mixed states and binary POVMs on local spaces of dimension at most d. Compactness gives an optimizer. Maximizing successively over the state and each effect replaces it, without changing dimension, by a pure state and six projections. Thus it suffices to treat a pure projective optimizer with deficit

$$\varepsilon = q_* - Q_d = \varepsilon_0 + \varepsilon_A + \varepsilon_B,$$

where the three terms are the expectations of the positive certificate remainders.

The proof uses the fixed rational constants

$$\mu = \frac{7}{8000}, \quad h_0 = 10^{-7}, \quad K = \frac{4656}{25}, \quad M = \frac{78}{5},$$

and

$$H = \frac{39}{10}K + \frac{\mu^2}{2}, \qquad \theta = \frac{\mu^2}{16H}, \qquad \Delta = \frac{\theta h_0}{160}.$$

The certified contact coordinate has Lipschitz distortion at most 20. Saturated quadratic coercivity gives the far-tail charge

$$m_{\mathrm{out}} \le C_{\mathrm{out}}\sqrt{\varepsilon_0}, \qquad C_{\mathrm{out}} = \frac{400\cdot 10^{12}}{1883^2}.$$

The near-fixed closure estimate, including all four source/target omissions, is

$$m_N \le C_N\sqrt{\varepsilon}, \qquad C_N = \frac{4}{\mu^2}\left(24 + 6K + HC_0\right), \qquad C_0 = 100\sqrt{40}\,h_0^{-1}.$$

On the retained drift sector use the shifted mesh $h_d = \Delta/(4\cdot 20^d)$. There are exactly $d+1$ principal and d intermediate frames, with full distortion sum

$$\sum_{k=0}^{d} 20^k + \sum_{k=0}^{d-1} 20^k = \frac{21\cdot 20^d - 2}{19}.$$

After contact, exit, and repeated near/far charges are included, the complete frame discard satisfies

$$F_d \le C_F 20^{2d}\sqrt{\varepsilon},$$

where

$$K_R = \frac{84\sqrt{40}}{19\Delta}, \qquad C_F = K_R + 3(C_{\mathrm{out}} + C_N).$$

The initial retained drift mass therefore obeys

$$W_D \ge 1 - C_I 20^{2d}\sqrt{\varepsilon}, \qquad C_I = C_{\mathrm{out}} + C_N + \frac{4\sqrt{40}}{\Delta}.$$

Canonical joint spectral packets make the intermediate target of one response literally the source of the next. At fixed time distinct packet labels are orthogonal. Along each retained chain, quantitative drift places successive local spectral projections in pairwise disjoint cells, so the chain length is at most d. No bound on the number of chains or the sum of their lengths is used. The full recurrence and terminal ledger gives

$$E_{\mathrm{exit}} + E_{\mathrm{rec}} \le Ad\varepsilon + BC_F 20^{2d}\sqrt{\varepsilon}, \qquad A = 5616, \quad B = \frac{200772}{25}.$$

The reverse endpoint estimate, summed over arbitrarily many chains, gives

$$E_{\mathrm{exit}} + E_{\mathrm{rec}} \ge \frac{W_D}{(d+1)M^{2d}}.$$

Define

$$\Gamma = (20M)^4 = 312^4 = 9{,}475{,}854{,}336$$

and

$$\kappa = \min\left\{1, \frac{1}{4C_I^2}, \frac{1}{8A}, \frac{1}{64B^2C_F^2}\right\}.$$

Exact arithmetic shows that the fourth entry is the minimum and

$$\kappa = 4.2946546143314459987533745197929408 51\ldots \times 10^{-52}.$$

If the loss in W_D is at least $1/2$, it directly gives the required exponential bound. Otherwise $W_D \ge 1/2$, and at least one of the two terms in the energy upper bound is half the recurrence lower bound. The first gives a stronger $d^{-2}M^{-2d}$ estimate; squaring the second and using $(d+1)^2 \le 4d^2$ gives, uniformly for every $d \ge 1$,

$$q_* - Q_d \ge \kappa d^{-4}\Gamma^{-d}.$$

The production proof is guarded at every local lemma and at the final algebra. A second reconstruction was preregistered against a sealed 19-source packet, rebuilt the complete mass and multiplicity ledger without access to the production assembly, and returned the same Γ. A third exact SymPy pass reconstructed A, B, Γ, and the minimum defining κ. The certificate retains the source snapshots and all three routes.

12 Reproduction

From the repository root, quick custody verification and full deterministic replay are respectively

```
python certificate/release/verify_release.py
python certificate/release/verify_release.py --full
```

The manifest and exact file-to-claim map are distributed beside this supplement.